

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1 #loading packages  
library(tidyverse)  
library(rvest)  
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2 #setting scraping website
webpage <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
webpage
```

```
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- Contact Person
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
#water system name
water_system <- webpage %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)")%>%
  html_text()
water_system
```

```
## [1] "Durham"
```

```
#contact person
contact_person <- webpage %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)")%>%
  html_text()
contact_person
```

```
## [1] "Reginald Hicks"
```

```
#ownership
ownership <- webpage %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)")%>%
  html_text()
ownership
```

```
## [1] "Municipality"
```

```
#Maximum Day Use (MGD) - for each month
```

```
max_MGD <- webpage %>%
  html_nodes("th~ td+ td") %>%
  html_text()
max_MGD
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

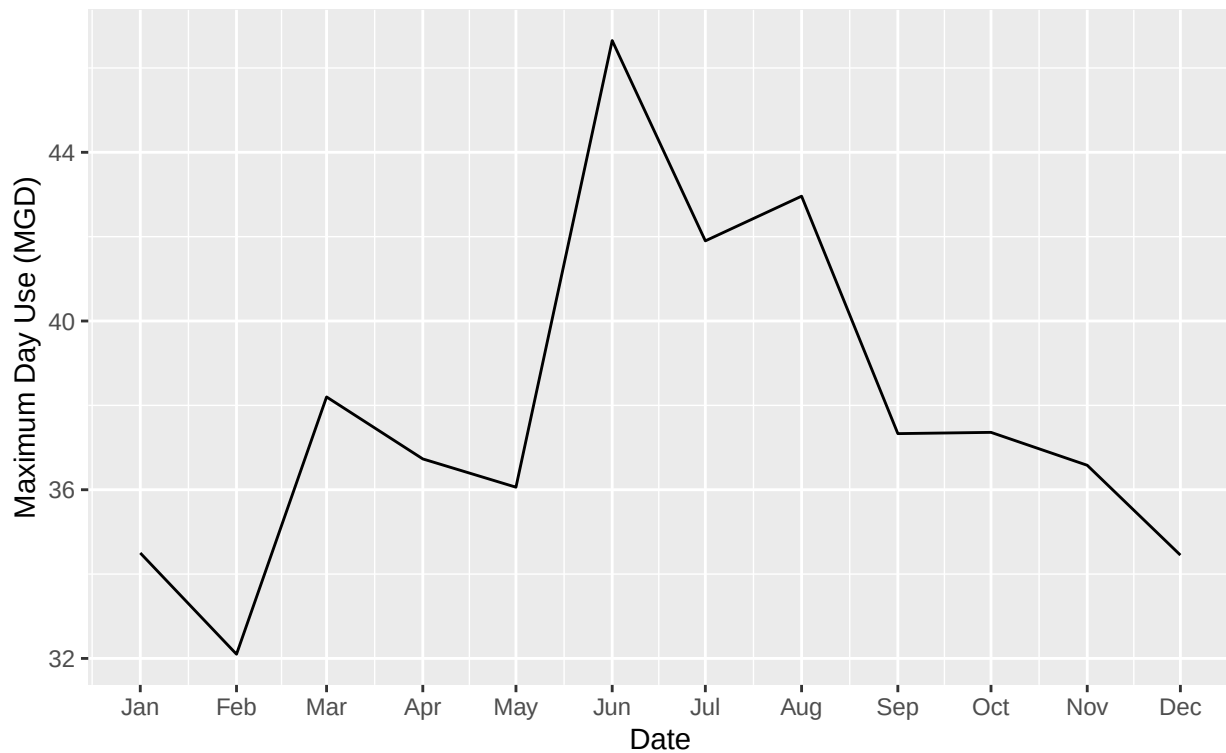
```
#4 #converting scraped data into a data frame
```

```
df_1 <- data.frame(
  "Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
  "Year" = rep(2024,12),
  "Monthly_MGD" = as.numeric(max_MGD) %>%
  mutate(Water_System = !!water_system,
         Ownership = !!ownership,
         Contact_Person = !!contact_person,
         Date = my(paste(Month,"-",Year)))
```

```
#5 #plotting MGD over time
```

```
df_1_plot <- ggplot(df_1,
  aes(x = Date, y = Monthly_MGD)) +
  geom_line() +
  labs(title = paste("2024 Water Usage Data for",
    water_system), subtitle = ownership,
    y = "Maximum Day Use (MGD)",
    x = "Date") +
  scale_x_date(date_breaks = "1 month",
    date_labels = "%b")
print(df_1_plot)
```

2024 Water Usage Data for Durham Municipality



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
#creating scraping function
scrape.it <- function(PWSID, the_year){

#retrieving the website contents
website <- read_html(paste0(
  'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=',
  PWSID, '&year=', the_year))

#setting elements address variables
water_system_tag <- 'div+ table tr:nth-child(1) td:nth-child(2)'
contact_person_tag <- 'div+ table tr:nth-child(4) td:nth-child(2)'
ownership_tag <- 'div+ table tr:nth-child(2) td:nth-child(4)'
mgd_tag <- 'th~ td+ td'

#scraping data
```

```

the_water_system <- website %>%
  html_nodes(water_system_tag) %>%
  html_text()
print(the_water_system)

the_contact_person <- website %>%
  html_nodes(contact_person_tag) %>%
  html_text()

the_ownership <- website %>%
  html_nodes(ownership_tag) %>%
  html_text()

the_mgd <- website %>%
  html_nodes(mgd_tag) %>%
  html_text()

#constructing data frame
data_frame <- data.frame(
  "Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
  "Year" = rep(the_year, 12), #year as function input
  "Max_Daily_MGD" = as.numeric(the_mgd) %>%
  mutate(Water_System = !!the_water_system,
         Ownership = !!the_ownership,
         Contact_Person = !!the_contact_person,
         PWSID = !!PWSID, #PWSID as function input
         Date = my(paste(Month,"-",Year)))

#scraping etiquette
Sys.sleep(1)

#returning data frame as output
return(data_frame)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7 #using the function for Durham 2020
df_2 <- scrape.it('03-32-010', 2020)

```

```
## [1] "Durham"
```

```

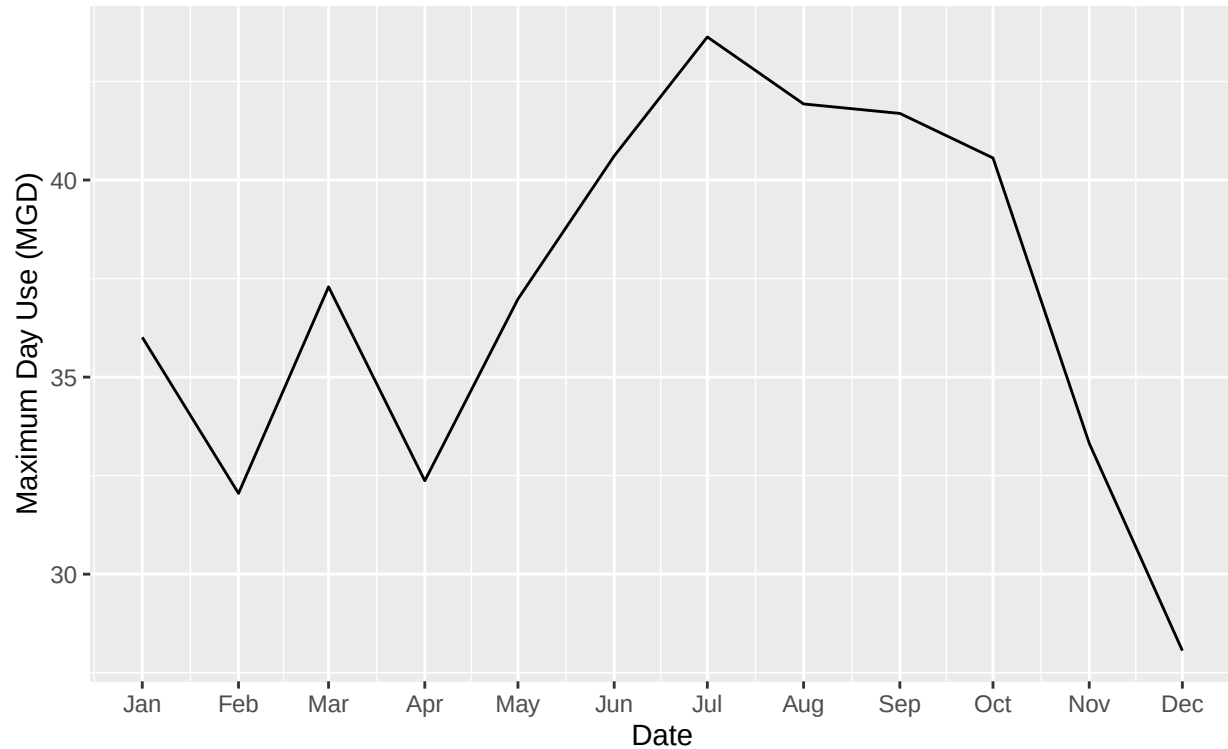
#plotting the results
df_2_plot <- ggplot(df_2,
  aes(x = Date, y = Max_Daily_MGD)) +
  geom_line() +
  labs(title = "2020 Water Usage Data for Durham",
       subtitle = "Municipality",
       y = "Maximum Day Use (MGD)",
       x = "Date") +

```

```
scale_x_date(date_breaks = "1 month",
             date_labels = "%b")

print(df_2_plot)
```

2020 Water Usage Data for Durham
Municipality



```
#showing the data frame
str(df_2)
```

```
## 'data.frame': 12 obs. of 8 variables:
## $ Month : num 1 5 9 2 6 10 3 7 11 4 ...
## $ Year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Max_Daily_MGD : num 36 37 41.7 32 40.6 ...
## $ Water_System : chr "Durham" "Durham" "Durham" "Durham" ...
## $ Ownership : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Contact_Person: chr "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
## $ PWSID : chr "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Date : Date, format: "2020-01-01" "2020-05-01" ...
```

- Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8 #using function for Cary 2020
df_3 <- scrape.it('03-92-020', 2020)
```

```
## [1] "Cary"
```

```
str(df_3)
```

```
## 'data.frame': 12 obs. of 8 variables:
## $ Month : num 1 5 9 2 6 10 3 7 11 4 ...
## $ Year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Max_Daily_MGD : num 21.9 23.2 25.5 17.7 26.2 ...
## $ Water_System : chr "Cary" "Cary" "Cary" "Cary" ...
## $ Ownership : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Contact_Person: chr "Jamie Revels" "Jamie Revels" "Jamie Revels" "Jamie Revels" ...
## $ PWSID : chr "03-92-020" "03-92-020" "03-92-020" "03-92-020" ...
## $ Date : Date, format: "2020-01-01" "2020-05-01" ...
```

```
#combining with Durham 2020 Data
```

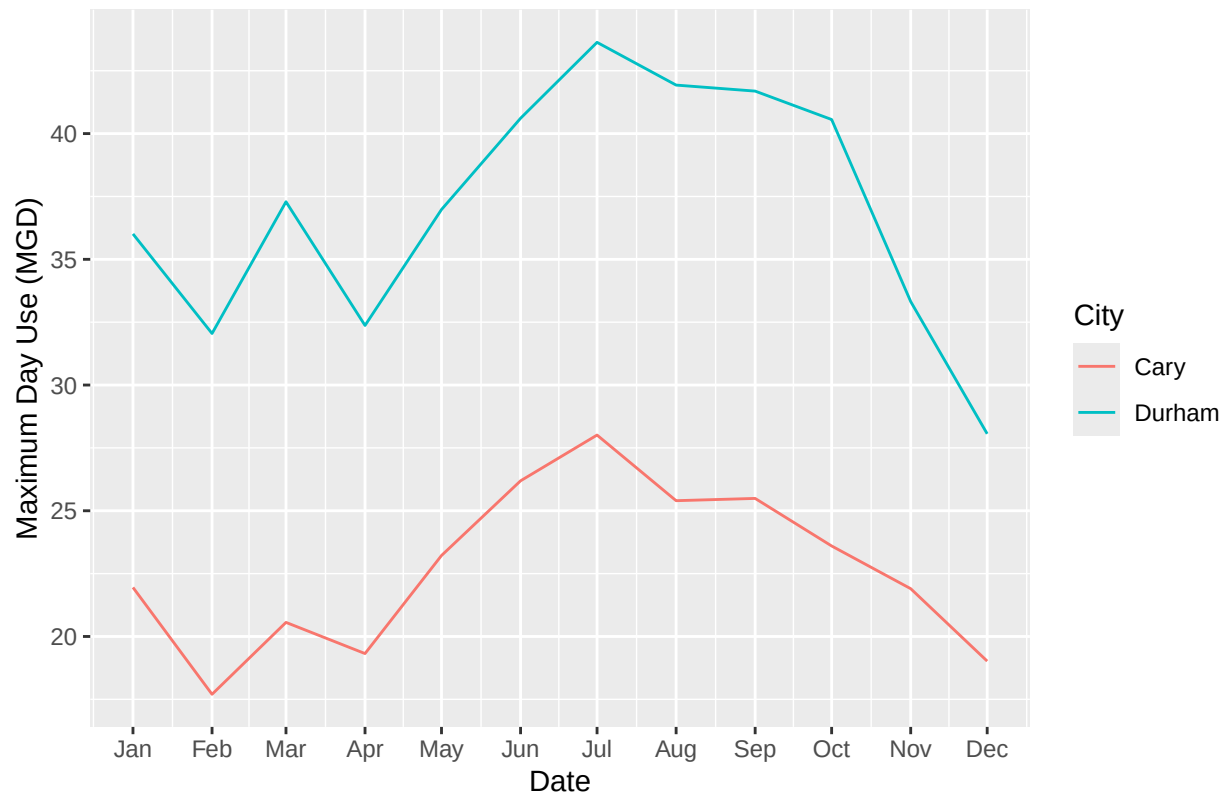
```
df_4 <- bind_rows(df_3, df_2)
```

```
#Plotting the comparison
```

```
durham_cary_plot <- ggplot(df_4,
  aes(x = Date, y = Max_Daily_MGD, color = Water_System)) +
  geom_line() +
  labs(title = "2020 Water Usage Data for Durham and Cary",
    color = "City",
    y = "Maximum Day Use (MGD)",
    x = "Date") +
  scale_x_date(date_breaks = "1 month",
    date_labels = "%b")
```

```
print(durham_cary_plot)
```

2020 Water Usage Data for Durham and Cary



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
#scraping years 2018 to 2023 - Cary
the_years = rep(2018:2023)
the_PWSID = '03-92-020'

#using purrr's to apply the scrape function
the_dfs <- map(the_years, scrape.it, PWSID=the_PWSID)
```

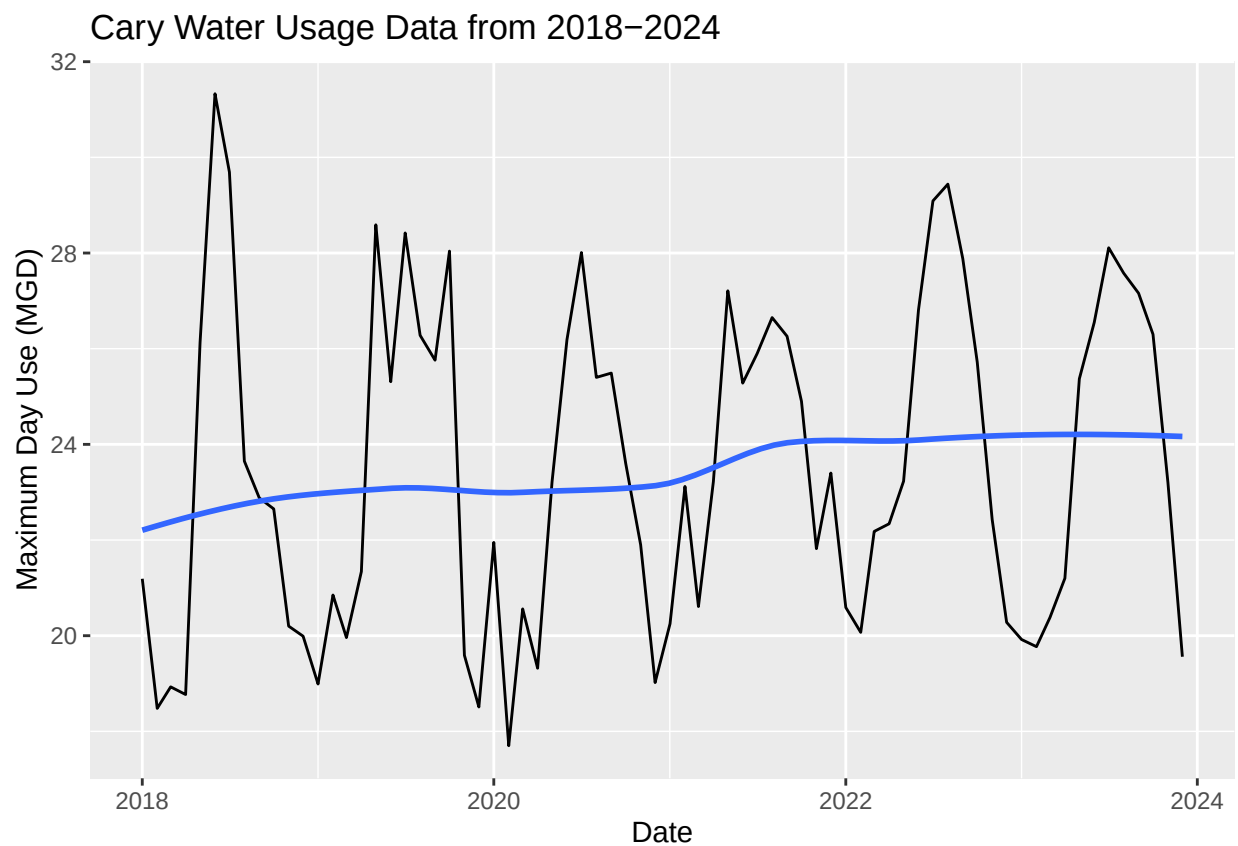
```
## [1] "Cary"
## [1] "Cary"
## [1] "Cary"
## [1] "Cary"
## [1] "Cary"
## [1] "Cary"
```

```
#combine to single data frame
df_5 <- bind_rows(the_dfs)

#plotting
cary_years_plot <- ggplot(df_5,
  aes(x = Date, y = Max_Daily_MGD)) +
  geom_line() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(title = "Cary Water Usage Data from 2018-2024",
    y="Maximum Day Use (MGD)",
    x="Date")

print(cary_years_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: It appears that water usage in Cary has been slightly increasing over time, however it is hard to know for sure whether the slight increase over a narrow time frame would be statistically significant.